

RAKESH TRIPATHY

Backend Engineer | Node.js | TypeScript | AI Backend | Distributed Systems

✉ rtripathy852@gmail.com ☎ +91-77354-64131 📍 Berhampur, Odisha, India 🔗 LinkedIn 🐙 GitHub 📁 Portfolio

SUMMARY

Backend Engineer with 3+ years of experience building scalable and secure backend applications using **Node.js, TypeScript, Express.js, Fastify, PostgreSQL, MongoDB, and Redis**. Skilled in developing **REST APIs, Microservices, JWT, OAuth2, RBAC, Distributed Systems, Caching, Docker, AWS, CI/CD, Clean Architecture, and System Design**. Experienced in building **AI-powered backend solutions** using **LLMs, RAG, LangChain, and Vector Databases** to deliver production-ready applications.

SKILLS

Languages

JavaScript (ES6+), TypeScript, SQL, Python

Databases

PostgreSQL, MongoDB, MySQL, Redis, Mongoose, Prisma, Sequelize

AI & GenAI

OpenAI API, LangChain, LangGraph, RAG, Vector Databases, ChromaDB, FAISS, Prompt Engineering

Cloud & DevOps

Docker, AWS EC2, AWS S3, Nginx, GitHub Actions, CI/CD

Testing & Documentation

Postman, Swagger/OpenAPI, Jest

Backend

Node.js, Express.js, Fastify, NestJS, REST APIs, GraphQL, Microservices

Authentication & Security

JWT, OAuth2, Role-Based Access Control (RBAC), API Security, Rate Limiting

Messaging & Background Jobs

Kafka, RabbitMQ, BullMQ

Architecture

System Design, Clean Architecture, SOLID Principles, Design Patterns, Distributed Systems, Event-Driven Architecture

PROFESSIONAL EXPERIENCE

Backend Engineer

MSol Technology

11/2024 – Present
Pune

- Developed scalable backend services using **Node.js, TypeScript, Fastify, PostgreSQL, MongoDB, and Redis**, delivering secure and high-performance REST APIs.
- Optimized database queries and implemented **Redis caching**, improving API response time by **40%** and reducing database load.
- Automated deployments using **Docker, AWS EC2, and GitHub Actions CI/CD**, enabling faster and reliable production releases.
- Collaborated with cross-functional teams to build production-ready backend systems following **Clean Architecture, SOLID principles, and API security**.

Backend Engineer

Telepie Technology

03/2023 – 10/2024
Delhi

- Built secure backend APIs using **Node.js, TypeScript, Express.js, and MongoDB** for scalable business applications.
- Implemented **JWT, OAuth2, and Role-Based Access Control (RBAC)** to strengthen authentication and authorization.
- Developed real-time features with **socket.io** and improved application performance through **MongoDB optimization and Redis caching**.
- Integrated **AWS S3** for secure file storage and collaborated with frontend and DevOps teams to deliver production-ready features.

PROJECTS

Scalable Multi-Vendor E-Commerce Backend

Node.js, TypeScript, Fastify, PostgreSQL, MongoDB, Redis, Docker, AWS EC2, GitHub Actions, JWT, RBAC, Kafka

- Designed and developed a production-ready multi-vendor backend using **Node.js, Fastify, PostgreSQL, MongoDB, and Clean Architecture**, supporting authentication, product, order, inventory, and payment modules.
- Implemented **JWT Authentication, RBAC, Redis caching, database indexing, pagination, and centralized error handling**, improving API performance and application security.
- Built asynchronous order processing using **Kafka**, integrated secure payment workflows, and exposed well-documented REST APIs with **Swagger/OpenAPI**.
- Containerized services with **Docker**, automated deployments using **GitHub Actions CI/CD**, and deployed applications on **AWS EC2** with production logging and monitoring.

AI-Powered Mental Wellness Platform

Node.js, TypeScript, MongoDB, OpenAI API, LangChain, ChromaDB, Redis, RabbitMQ, Socket.io, JWT

- Developed scalable backend APIs for therapist, patient, appointment, chat, and session management using **Node.js, TypeScript, MongoDB, and Express/Fastify**.
- Integrated **OpenAI API, LangChain, RAG, and ChromaDB** to deliver AI-powered recommendations and intelligent conversation capabilities.
- Implemented **JWT Authentication, RBAC, Redis caching, RabbitMQ background jobs, and socket.io** for secure access, real-time communication, and asynchronous processing.
- Optimized backend architecture with **Clean Architecture, API validation, centralized logging, Docker, and CI/CD**, ensuring secure and scalable production deployment.

EDUCATION

Bachelor of Technology (B.Tech)

BPUT

Electronics & Telecommunication Engineering

07/2018 – 08/2022
Odisha